

DEPARTMENT OF

**COMPUTER
SCIENCE AND
ENGINEERING**

VISION OF THE DEPARTMENT

To achieve academic and research excellence in essential technologies of Computer Science and Engineering by promoting a creative environment for learning and innovation.

MISSION OF THE DEPARTMENT

- To provide dynamic, innovative and flexible curriculum which equip the students with the necessary problem driven skills to strengthen their career prospects and potential to pursue higher studies.
- To foster inquisitive-driven research among students and staff so as to reinforce the domain knowledge and address contemporary societal issues.
- To inculcate ethical values, leadership qualities and professional behaviour skills for improving the living standards of people

B. TECH.
(COMPUTER SCIENCE AND
BUSINESS SYSTEMS)

B. TECH. COMPUTER SCIENCE AND BUSINESS SYSTEMS

PROGRAM EDUCATIONAL OBJECTIVES

PEO-I: To equip students with the essential knowledge, skillsets and attitude to be society and industry ready and make a meaningful contribution to the growth of the Indian economy.

PEO-II: To prepare students with fundamental concepts of Computer Science and additionally develop an inherent capacity for liberal arts, innovative mind-set, life values and an appreciation of sustainability issues.

PEO-III: To prepare 'Business Engineers', a cluster of engineering talent tuned to the needs of Business 4.0.

PEO-IV: To enable students to appreciate the technologies of the future and understand the fundamental concepts of business management and develop the 'Innovation' mind-set.

PEO-V: To ensure students are future-ready with emerging topics such as Cyber Security, Machine Learning, Cloud Computing, IoT, Analytics etc. as part of the curriculum.

B. TECH. COMPUTER SCIENCE AND BUSINESS SYSTEMS

PROGRAM OUTCOMES

PO-1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO-2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4).

PO-3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5).

PO-4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO-5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including

prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6).

PO-6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7)

PO-7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9).

PO-8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO-9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

PO-10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO-11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii)

adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8).

B. TECH. COMPUTER SCIENCE AND BUSINESS SYSTEMS

PROGRAM SPECIFIC OUTCOMES

PSO-1: Apply the fundamentals of mathematics, science and technology abstraction to develop computational tools and applications in the areas related to algorithms, big data analytics, machine learning, and artificial intelligence and networking.

PSO-2: Leverage new age technologies to solve contemporary challenges of varying complexity and develop 'Innovation' mind-set and entrepreneurial skills.

PSO-3: Be adept at designing business solutions using the fundamental concepts of business management, leadership and entrepreneurial skills and appreciate life values and sustainability issues.

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD

B. TECH. I YEAR

COMPUTER SCIENCE AND BUSINESS SYSTEMS

I SEMESTER

R25

Course Code	Title of the Course	L	T	P/D	CH	C
25BS1MT102	Discrete Mathematics	2	1	0	3	3
25BS1MT103	Introductory Topics in Statistics, Probability and Calculus	2	1	0	3	3
25ES1CB101	Fundamentals of Computer Science	3	0	0	3	3
25ES1EE106	Principles of Electrical Engineering	3	0	0	3	3
25BS1PH103	Physics for Computing Science	3	0	0	3	3
25ES2CB101	Fundamentals of Computer Science Laboratory	0	0	2	2	1
25ES2EE106	Principles of Electrical Engineering Laboratory	0	0	2	2	1
25BS2PH103	Physics for Computing Science Laboratory	0	0	2	2	1
25HS2EN102	Business Communication & Value Science – I	1	0	2	3	2
25MN6HS101	Induction Programme	2	0	0	2	0
Total		16	2	8	26	20

II SEMESTER

R25

Course Code	Title of the Course	L	T	P/D	CH	C
25BS1MT107	Linear Algebra	2	1	0	3	3
25BS1MT108	Statistical Modelling	2	1	0	3	3
25PC1CB101	Data Structures and Algorithms	3	0	0	3	3
25ES1EC102	Principles of Electronics Engineering	3	0	0	3	3
25HS1MG101	Fundamentals of Management	3	0	0	3	3
25PC2CB101	Data Structures and Algorithms Laboratory	0	0	2	2	1
25ES2EC102	Principles of Electronics Engineering Laboratory	0	0	2	2	1
25PC2CB102	Python Programming	0	0	2	2	1
25HS2EN103	Business Communication & Value Science – II	1	0	2	3	2
25MN6HS102	Environmental Science	2	0	0	2	0
Total		16	2	8	26	20

L – Lecture T – Tutorial P – Practical D – Drawing
 C – Credits SE – Sessional Examination CA – Class Assessment
 SEE – Semester End Examination D-D – Day to Day Evaluation
 CP – Course Project PE – Practical Examination

CH – Contact Hours/Week
 ELA – Experiential Learning Assessment
 LR – Lab Record

B. Tech. I Semester

(25BS1MT102) DISCRETE MATHEMATICS

TEACHING SCHEME		
L	T/P	C
2	1	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Set, Relation, Mapping, Algebraic Equations

COURSE OBJECTIVES:

- To know concepts of sets and relation to understand Group's and Ring theory
- To generate function, recurrence relations and it's solutions
- To learn boolean expressions, operations and truth tables
- To know graphs as models for many standard problems

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Represent characteristics of Sets, Groups, Rings and Fields

CO-2: Explain and exemplify tautology, contradiction and contingency

CO-3: Create generating function and solve the recurrence relations

CO-4: Analyse the design of various combinational & sequential logic circuits using the concepts of Boolean Algebra

CO-5: Apply graph theory based tools in solving practical problems

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	1	2	-	-	-	-	-	-	3	1	2
CO-2	3	3	2	2	-	-	-	-	-	-	-	2	1	1
CO-3	3	2	3	2	-	-	-	-	-	-	-	3	1	2
CO-4	3	2	3	3	2	-	-	-	-	-	-	3	1	2
CO-5	3	2	3	2	3	-	-	-	-	-	-	3	1	1

UNIT-I:

Abstract Algebra: Sets, Finite sets, Power sets, Set Operations, and duality, Relations, Types of relations, Equivalence relations, Partial Ordering, Partition of a set, Groups, subgroups, Lagrange's theorem on finite groups (without proof). Introduction to Ring, Integral domain and Field.

UNIT-II:

Logic: Propositional calculus - propositions and connectives, Conditional and Bi-conditional statements, truth assignments and truth tables, validity and satisfiability, tautology, contradiction, and contingency. Logical Equivalence and normal forms; Algebra of propositions, Logical implication, Quantifiers, Negation of quantified statements.

UNIT-III:

Recurrence Relations: Introduction, sequences, generating functions, coefficients of generating functions, recurrence relations, solutions of homogeneous and non-homogeneous recurrence relations using characteristic roots method.

UNIT-IV:

Boolean Algebra: Introduction of Boolean algebra, principle of duality, Basic logic gates, truth table, Boolean expressions, canonical form, Karnaugh map.

UNIT-V:

Graph Theory: Graphs, Types of Graphs, digraphs, adjacency matrix, incidence matrix, isomorphism, Trees, Properties of trees, Spanning trees, Minimal Spanning trees using Kruskal's and Prim's Algorithms. Eulerian paths and circuits in graphs and digraphs, Hamiltonian paths and circuits in graphs, Planar graphs, dual of a planar graph, Euler's formula, chromatic number, statement of Four-color theorem.

TEXT BOOKS:

1. Topics in Algebra, I. N. Herstein, 2nd Edition, John Wiley and Sons, 1975
2. Digital Logic & Computer Design, M. Morris Mano, 2nd Edition, Pearson, 2017
3. Elements of Discrete Mathematics, C. L. Liu, 2nd Edition, McGraw-Hill, 1985
4. Graph Theory with Applications, J. A. Bondy and U. S. R. Murty, 2nd Edition, Macmillan Press, London, 1978

REFERENCES:

1. Introduction to Linear Algebra, Gilbert Strang, 5th Edition, Wellesley, 2017
2. Introductory Combinatorics, R. A. Brualdi, North-Holland, New York, 3rd Edition, Prentice Hall, 1998
3. Graph Theory with Applications to Engineering and Computer Science, N. Deo, Prentice Hall, Englewood Cliffs, 1974
4. Introduction to Mathematical Logic, E. Mendelsohn, Van-Nostrand, 2nd Edition, London
5. Mathematical Logic for Computer Science, L. Zhongwan, 2nd Edition, World Scientific, 1998

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25BS1MT103) INTRODUCTORY TOPICS IN STATISTICS, PROBABILITY AND CALCULUS

TEACHING SCHEME		
L	T/P	C
2	1	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Permutations, Combinations and Basic Calculus

COURSE OBJECTIVES:

- To learn basic probability theory and statistical parameters
- To know the different types of probability distributions
- To learn basic objectives of statistic, classification and descriptive measures of data
- To know basic optimization techniques
- To find the differential and integral calculus

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Compute expected values of discrete and continuous random variables

CO-2: Identify the suitable probability distribution to solve the problems

CO-3: Classify and analyze the given data through basic statistics

CO-4: Apply basic optimization techniques to problems involving functions of two variables

CO-5: Calculate areas and volumes of solids by applying multiple integrals

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	2	2	-	-	-	-	1	-	2	1	1
CO-2	3	3	2	2	2	-	-	-	-	1	-	3	1	1
CO-3	3	2	2	2	2	-	-	1	1	2	-	3	1	2
CO-4	3	3	3	2	2	-	1	-	-	1	1	3	1	1
CO-5	3	3	3	2	3	-	1	-	-	1	1	3	1	1

UNIT-I:

Basic Probability & Mathematical Expectations: Concept of experiments, sample space, event, Definition of Combinatorial Probability, Conditional Probability, Baye's Theorem. Discrete and continuous random variables

Expected Values and Moments: mathematical expectation and its properties, Moments (including variance) and their properties, Moment generating function.

UNIT-II:

Probability Distributions

Discrete Distributions: Binomial, Poisson and Geometric distribution.

Continuous Distributions: Uniform, Exponential, Normal distributions.

UNIT-III:

Introduction to Statistics: Definition of Statistics. Basic objectives. Applications in various branches of science with examples.

Collection of Data: Internal and external data, Primary and secondary Data. Population and sample, Representative sample.

Descriptive Statistics: Classification and tabulation of univariate data, graphical representation, Frequency curves. Descriptive measures - central tendency and dispersion. Bivariate data. Summarization, marginal and conditional frequency distribution.

UNIT-IV:

Differential Calculus: Function of two variables- Limit, continuity, partial derivatives, Taylor's and Maclaurin's series expansions, Maxima and minima of function of two variables, Functions of three variables - Lagrange's method of undetermined multipliers.

UNIT-V:

Integral Calculus: Multiple Integrals- double integrals with constant and variable limits (Cartesian and polar form), change of order of integration (Cartesian form), triple integrals (Cartesian coordinates), applications of double and triple integrals: Area as double integration in Cartesian coordinates and Volume as a triple integration.

TEXT BOOKS:

1. Introduction of Probability Models, S. M. Ross, 11th edition, Academic Press, N. Y., 2014
2. Introduction to Probability and Statistics for Engineers and Scientists, Sheldon M. Ross, 4th Edition, Academic Press, 2009
3. Fundamentals of Statistics, A. Goon, M. Gupta and B. Dasgupta, Vol. I & II, World Press

REFERENCES:

1. Probability and Statistics for Engineers, I. R. Miller, J. E. Freund and R. Johnson, 4th Edition, PHI
2. Introduction to the Theory of Statistics, A. M. Mood, F. A. Graybill and D. C. Boes, McGraw-Hill Education
3. Advanced Engineering Mathematics, Peter V. O'Neil, 7th edition, Thomson learning
4. Advanced Engineering Mathematics, M. D. Greenberg, 2nd Edition, Pearson Education
5. Applied Mathematics, P. N. Wartikar and J. N. Wartikar, Vol. I & II, Vidyarthi Prakashan

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25ES1CB101) FUNDAMENTALS OF COMPUTER SCIENCE

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE OBJECTIVES:

- To introduce problem-solving through algorithms, flowcharts, operators, and variable naming conventions
- To develop the ability to apply structured control flow constructs for effective program design
- To understand functions, recursion, scope, and standard library usage in program development
- To explore pointers, arrays, and structures for efficient data manipulation and modular programming
- To apply dynamic memory allocation and Linux system-level file operations for real-world programming tasks

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Design algorithms and flowcharts and Apply operators and expressions with proper variable naming conventions

CO-2: Implement structured control flow constructs and differentiate them from unstructured approaches

CO-3: Develop modular programs using functions, recursion, and Analyze scope and lifetime of variables with standard library usage

CO-4: Manipulate data using pointers, arrays, and structures, and Evaluate solutions using pointer-based function arguments

CO-5: Implement dynamic memory allocation, Handle file operations in Linux, and Create modular programs using user-defined libraries and utilities

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	-	-	-	-	-	-	-	-	-	3	-	-
CO-2	3	3	2	-	-	-	-	-	-	-	-	3	-	-
CO-3	3	2	2	-	-	-	-	-	-	-	2	3	2	-
CO-4	3	2	2	-	2	-	-	-	-	-	-	3	2	-
CO-5	3	-	2	-	3	-	-	-	-	2	2	3	2	-

UNIT-I:

General Problem Solving Concepts: Algorithm, and Flowchart for problem solving with Sequential Logic Structure, Decisions and Loops. Operator and Expressions with discussion of variable naming and Hungarian Notation: Variable Names, Data Type and Sizes (Little Endian Big Endian), Constants, Declarations, Arithmetic Operators, Relational Operators, Logical Operators, Type Conversion, Increment Decrement Operators, Bitwise Operators, Assignment Operators and Expressions, Precedence and Order of Evaluation, proper variable naming and Hungarian Notation.

UNIT-II:

Control Flow with Discussion on Structured and Unstructured Programming: Statements and Blocks, If-Else-If, Switch, Loops – while, do, for, break and continue, Goto Labels, structured and un- structured programming.

UNIT-III:

Functions and Program Structure with Discussion on Standard Library: Basics of functions, parameter passing and returning type, C main return as integer, External, Auto, Local, Static, Register Variables, Scope Rules, Block structure, Initialization, Recursion, Preprocessor, Standard Library Functions and return types.

UNIT-IV:

Pointers, Arrays and Structures: Pointers and address, Pointers and Function Arguments, Pointers and Arrays, Address Arithmetic, character Pointers and Functions, Pointer Arrays, Pointer to Pointer, Initialization of Pointer Arrays, Functions returning pointers, command line Arguments, Pointers to functions. Basic Structures, Structures and Functions, Array of structures, Pointer of structures, typedef, Unions, Bit-fields.

UNIT-V:

Dynamic Memory Allocation: malloc, calloc, realloc, free. Input and Output, Linux System Interface, Programming Method: Standard I/O, Formatted Output – printf, Formatted Input – scanf, Variable length argument list, file access including FILE structure, fopen, stdin, stdout and stderr, Error Handling , Low level I/O – read and write, Open, create, close and unlink, Random access – lseek, Discussions on Listing Directory, Macro, User Defined Header, User Defined Library Function, Make File Utility.

TEXT BOOKS:

1. The C Programming Language, B. W. Kernighan and D. M. Ritchi, 2nd Edition, PHI
2. Programming in C, B. Gottfried, 2nd Edition, Schaum Outline Series
3. Let Us C, Yashavant P. Kanetkar, 16th Edition, 2019, BPB Publications

REFERENCES:

1. C: The Complete Reference, Herbert Schildt, 4th Edition, McGraw-Hill
2. Problem Solving with C, Jacqueline A. Jones and Keith Harrow, Pearson Education
3. Programming in ANSI C, E. Balguruswamy, 8th Edition, McGraw-Hill, 2019
4. A Book on C, Al Kelley, Ira Pohl, 4th Edition, Pearson Education
5. Stephen G. Kochan, Programming in C, CBS Publishers

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25ES1EE106) PRINCIPLES OF ELECTRICAL ENGINEERING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Mathematics, Physics

COURSE OBJECTIVES:

- To understand the basic concepts of electrical and magnetic circuits
- To understand the electromechanical energy conversion process in machines
- To identify the types of sensors and measure quantities in AC and DC systems
- To study various electrical installation components and safety measures

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the basic concepts and terminology of electrical quantities

CO-2: Analyze the DC circuit using various network theorems

CO-3: Analyze the electrical parameters of AC circuits with R-L-C elements

CO-4: Understand the concepts of Electrostatic, Electromagnetic fields, and operation of Electrical Machines

CO-5: Understand the components of Low Voltage Electrical Installations and the safety measures

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	1	2	3	3	2	-	-	-	1	-	3	2	1
CO-2	3	3	3	3	2	2	-	1	-	-	-	3	2	1
CO-3	3	2	3	3	2	3	-	-	-	-	-	3	1	1
CO-4	3	3	2	3	3	2	-	-	-	1	-	3	2	1
CO-5	2	2	3	2	3	3	-	-	-	-	-	3	2	2

UNIT-I:

Basic Circuit Concepts and Theorems: Concept of voltage, potential difference, current- Fundamental linear passive and active elements to their functional current voltage relation, independent voltage source and Independent current sources: ideal and practical sources, Kirchhoff's laws-series and parallel connections-Network solutions using mesh and nodal analysis, Theorems: Thevenin's and Norton's theorems, Superposition theorem.

UNIT-II:

AC Circuit Analysis: Representation of Sinusoidal waveform- RMS and average values, form factor and peak factor-series RL, RC and RLC circuits -phasor representation in polar and rectangular form, concept of impedance, admittance - active, reactive and apparent powers, power factor-Three phase balanced circuits: Star and delta connections (Derivation Only).

UNIT-III:

Electro Static and Electro Magnetic Fields:

Electrostatic Fields: Electric field intensity and strength, energy stored in a capacitor, charging, and discharging of capacitors absolute and relative permittivities.

Electromagnetic Fields: Magneto Motive Force, Flux Density, Faraday's laws of Electromagnetic Induction, self and mutual inductances, series magnetic circuit, magnetic materials and BH curve.

UNIT-IV:

Transformers and DC Machines:

Transformer: Single phase transformer principle, emf equation, Transformation ratio, ideal and practical transformer, losses in a transformer, Efficiency and Voltage regulation (Elementary Treatment only)

DC Machines: DC generator construction, principle, emf generated, types of DC generators, DC motor principle, back emf.

UNIT-V:

Measurements, Electrical Installation and Safety Measures: Introduction to electrical measurements, types of instruments, indicating type instruments (MC and MI) (Elementary treatment only)

Electrical Installation and Safety Measures:

Components of LT Switchgear: Switch Fuse Unit (SFU), MCB, MCCB, Types of Wiring, Earthing: Need of earthing, Types (Pipe and Plate earthing), classification of batteries, working and electrical characteristics of Lead Acid battery.

TEXT BOOKS:

1. Basic Electrical Engineering, D. C. Kulshreshtha, 2nd Edition, TMH, 2019
2. Basic of Electrical Engineering, T. K. Nagsarkar and M. S. Sukhija, Oxford University Press, 2011
3. Electromagnetic Field Theory, K. A. Gangadhar, P. M. Ramanathan, 16th Edition, Khanna, 2011

REFERENCES:

1. Basic Electrical Engineering, V. K. Mehta, S. Chand and Company
2. Circuits and Networks Analysis and Synthesis, A. Sudhakar, Shyammmohan S. Palli, Tata McGraw-Hill, 2010
3. Engineering Electromagnetics, William H. Hayt, Jr. John A. Buck, 8th Revised Edition, McGraw-Hill Higher, 2011
4. Fundamentals of Electrical and Electronics Engineering, Smarjith Ghosh, Prentice Hall, 2010

ONLINE RESOURCES:

1. <https://nptel.ac.in/courses/117106108>

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25BS1PH103) PHYSICS FOR COMPUTING SCIENCE

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: 10 + 2 Physics

COURSE OBJECTIVES:

- To discuss the fundamentals of oscillatory systems
- To analyze various phenomena of light- Interference and Diffraction
- To apply the basic principles of LASER to various laser systems and optical fibers
- To explain the basic concepts in quantum physics and semiconductors
- To understand the crystal structures and state the laws of thermodynamics

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Describe the fundamentals of oscillatory systems

CO-2: Extend the importance of Interference in thin films and diffraction

CO-3: Explain the lasing action of various laser sources and optical fiber materials

CO-4: Apply the concepts of quantum mechanics and semiconductors for engineering applications

CO-5: Recall the importance crystal structures and applications of thermodynamics

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	-	-	-	1	-	-	-	-	1	-	-	-
CO-2	3	2	-	1	-	1	-	-	-	1	1	-	-	-
CO-3	3	2	1	1	-	1	-	-	-	-	1	1	-	-
CO-4	3	2	1	1	-	1	-	-	-	1	1	1	-	-
CO-5	3	2	-	1	-	1	-	-	-	-	1	-	-	-

UNIT-I:

Oscillations: Periodic motion-simple harmonic motion-characteristics of simple harmonic motion-vibration of simple spring- mass system. Resonance-definition., damped harmonic oscillator – heavy, critical and light damping, energy decay in a damped harmonic oscillator, quality factor.

UNIT-II:

Wave Optics: Interference-principle of superposition-young's experiment, Temporal and Spatial Coherence, Theory of interference fringes-types of interference-Fresnel's prism-Newton's rings, Diffraction- Two kinds of diffraction-Difference between interference and diffraction- Fraunhofer diffraction at single slit-plane diffraction grating.

UNIT-III:

Laser and Fiber Optics: Properties of laser beams: mono-chromaticity, coherence, directionality and brightness, Einstein's theory of matter radiation interaction and A and B coefficients; amplification of light by population inversion, different types of lasers: Ruby Laser, CO₂ applications of lasers in engineering. Fiber optics, Types of optical fibers and Applications.

UNIT-IV:

Quantum Mechanics: Introduction - Planck's quantum theory- Matter waves, de-Broglie wavelength, Heisenberg's Uncertainty principle, time independent Schrödinger's wave equation, Physical significance of wave function, Particle in a one-dimensional potential box.

Semiconductor Physics: Basic concept of Band theory; conductor, semiconductor and Insulator.

UNIT-V:

Crystallography: Basic terms-types of crystal systems, Bravais lattices, miller indices, d spacing, Atomic packing factor for SC, BCC, and FCC.

Thermodynamics: Zeroth law of thermodynamics, first law of thermodynamics, brief discussion on application of 1st law, second law of thermodynamics, entropy, change in entropy in reversible and irreversible processes.

TEXT BOOKS:

1. Concepts of Modern Physics, Beiser A., 5th Edition, McGraw-Hill International
2. Fundamentals of Physics, David Halliday, Robert Resnick, Jearl Walker, Wiley
3. Physics for Computer Science Students: With Emphasis on Atomic and Semiconductor Physics, Damask, A., Garcia, N., Springer, 2012

REFERENCES:

1. Optics, Ajoy Ghatak, 5th Edition, Tata McGraw-Hill
2. University Physics, Sears & Zemansky, Addison-Wesley
3. Fundamentals of Optics, Jenkins and White, 3rd Edition, McGraw-Hill
4. College Physics, Urone P. P., Brooks/Cole, 1997

B. Tech. I Semester

(25ES2CB101) FUNDAMENTALS OF COMPUTER SCIENCE LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To develop problem-solving skills through algorithms, flowcharts, and structured programming
- To provide practical exposure to operators, control structures, loops, and decision-making constructs
- To enable students to design modular programs using functions, arrays, storage classes, and structures
- To enhance the ability to handle advanced programming concepts like pointers, string manipulation, and dynamic memory allocation
- To familiarize students with Linux-based file handling and system-level programming for real-world applications

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Design and Implement algorithms, flowcharts, and control structures to solve computational problems

CO-2: Develop modular programs using functions, arrays, structures, unions, and bitwise operations

CO-3: Apply string manipulation techniques and pointers for solving advanced problems

CO-4: Implement substring matching, searching algorithms, and dynamic memory management effectively

CO-5: Perform file operations using Linux system calls and standard I/O functions

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	-	-	-	-	-	-	-	-	-	3	-	-
CO-2	3	2	2	-	-	-	-	-	-	-	-	3	-	-
CO-3	3	2	-	-	2	-	-	-	-	-	-	3	-	-
CO-4	3	2	2	-	2	-	-	-	-	-	-	3	2	-
CO-5	3	-	2	-	2	-	-	-	-	-	2	3	2	-

LIST OF EXPERIMENTS:

WEEK 1: Algorithm and flowcharts of small problems like GCD Structured code writing

WEEK 2: Small but tricky codes on Operators and Decision Making

WEEK 3: Tricky Codes in Loops and Nested Loops

WEEK 4: Functions, Storage Classes

WEEK 5: Arrays

WEEK 6: Structures, Unions, Bit Manipulation

WEEK 7: String Manipulation

WEEK 8: Pointers, advanced functions

WEEK 9: Interesting substring matching / searching programs

WEEK 10: Dynamic Memory Allocation

WEEK 11 & WEEK 12 & WEEK 13: File handling using Linux

TEXT BOOKS:

1. The C Programming Language, B. W. Kernighan and D. M. Ritchi, 2nd Edition, PHI
2. Programming in C, B. Gottfried, 2nd Edition, Schaum Outline Series
3. Let Us C, Yashwant P. Kanetkar, 16th Edition, 2019, BPB Publications

REFERENCES:

1. C: The Complete Reference, Herbert Schildt, 4th Edition, McGraw-Hill
2. Problem Solving with C, Jacqueline A. Jones and Keith Harrow, Pearson Education
3. Programming in ANSI C, E. Balguruswamy, 8th Edition, McGraw-Hill, 2019
4. A Book on C, Al Kelley, Ira Pohl, 4th Edition, Pearson Education
5. Stephen G. Kochan, Programming in C, CBS Publishers

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25ES2EE106) PRINCIPLES OF ELECTRICAL ENGINEERING LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: Principles of Electrical Engineering

COURSE OBJECTIVES:

- To design electrical systems
- To analyse a given network by applying various network theorems
- To verify phase relationships in star and delta connected three phase networks
- To study various electrical safety precautions

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the basic concepts and terminology of electrical quantities

CO-2: Analyse the DC circuits using various network theorems

CO-3: Analyse the electrical parameters of AC circuits with R-L-C elements

CO-4: Control different electrical machines and Transformers and evaluate their performance using different testing methods

CO-5: Simulate the electrical circuits using suitable software

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	1	2	3	3	2	-	-	-	1	-	3	2	1
CO-2	3	3	3	3	2	2	-	1	-	-	-	3	2	1
CO-3	3	2	3	3	2	3	-	-	-	-	-	3	1	1
CO-4	3	3	2	3	3	2	-	-	-	1	-	3	2	1
CO-5	3	3	2	3	3	2	-	-	-	-	-	3	3	3

LIST OF EXPERIMENTS:

1. Demonstration of safety precautions, measuring instruments and electrical installation components
2. Verification of KVL and KCL
3. Verification of Thevenin's theorem
4. Verification of Norton's theorem
5. Verification of Superposition theorem
6. Analysis of single-phase RL, RC and RLC series circuits
7. Load Test on Single Phase Transformer (Calculate Efficiency and Regulation)
8. Verification of voltage and current relationships in a star connected balanced three Phase Circuit
9. Measurement of electrical quantities in DC and AC circuits
10. Open circuit characteristics of separately excited DC generator

11. Simulation of series RLC circuit ($X_L > X_C$, and $X_L < X_C$)
12. Simulation of time response of RC circuit

TEXT BOOKS:

1. Basic Electrical Engineering, D. P. Kothari and I. J. Nagrath, 4th Edition, Tata McGraw-Hill, 2019
2. Basic Electrical Engineering, M. S. Naidu and S. Kamakshaiah, 2nd Edition, Tata McGraw-Hill, 2008

REFERENCES:

1. Basic Electrical Engineering, P. Ramana, M. Suryakalavathi, G. T. Chandrashekar, 2nd Edition, S. Chand, 2019
2. Basic Electrical Engineering, D. C. Kulshreshtha, McGraw-Hill, 2009
3. Basic Electrical and Electronics Engineering, M. S. Sukhija, T. K. Nagsarkar, 1st Edition, Oxford, 2012
4. Basic Electrical Engineering, Abhijit Chakrabarthy, Sudipta Debnath, Chandan Kumar Chanda, 2nd Edition, McGraw-Hill, 2021
5. Electrical and Electronics Technology, E. Hughes, Pearson, 2010

ONLINE RESOURCE:

1. <https://nptel.ac.in/courses/117106108>

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25BS2PH103) PHYSICS FOR COMPUTING SCIENCE LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: 10+2 Physics

COURSE OBJECTIVES:

- To verify Biot –Savart law
- To study Hall co-efficient of semiconductor
- To determine Planck's and Stefan's constant
- To practically learn interaction of light with matter through physical phenomena like interference, diffraction and total internal reflection
- To compare the experimental results with the classroom learning

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Realize tangent law of magnetism

CO-2: Evaluate the hall coefficient of given semi conductor

CO-3: Evaluate the Planck's and Stefan's constant

CO-4: Demonstrate the optical phenomena with formation of Newton Rings and to evaluate grating parameters

CO-5: Correlate the experimental results with the classroom learning

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	3	1	2	1	2	1	3	2	-	1	1	-	-
CO-2	3	3	1	2	1	2	1	3	2	-	1	1	-	-
CO-3	3	3	1	2	1	2	1	3	2	-	1	1	-	-
CO-4	3	3	1	2	1	2	1	3	2	-	1	1	-	-
CO-5	3	3	1	2	2	2	1	3	3	2	1	-	2	-

LIST OF EXPERIMENTS:

1. Magnetic field along the axis of current carrying coil – Stewart and Gee
2. Determination of Hall coefficient of semi-conductor
3. Determination of Plank constant
4. Determination of wavelength of light by Laser diffraction method
5. Determination of wavelength of light by Newton's Ring method
6. Determination of laser and optical fiber parameters
7. Determination of Stefan's Constant

TEXT BOOKS:

1. Physics for Computing Science Laboratory Manual/Observation, Physics Faculty of VNRVJIE
2. A Textbook of Practical Physics, S. Balasubramanian, M. N. Srinivasan, S Chand Publishers, 2017

ONLINE RESOURCES:

1. <https://vlab.amrita.edu/index.php?sub=1&brch=189&sim=343&cnt=1>
2. <https://vlab.amrita.edu/index.php?sub=1&brch=280&sim=1518&cnt=1>

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25HS2EN102) BUSINESS COMMUNICATION AND VALUE SCIENCE - I

TEACHING SCHEME		
L	T/P	C
0	2	2

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES:

- Basic communication in tenses (past, present, future)
- Awareness of common words (adjectives used in daily verbal communication)
- Basic idea of sentence formation and thereby paragraph building and writing
- Communication according to daily and varied contextual scenarios
- Basic communication model/channel (sender, receiver, and feedback), Active and passive listening skills
- Basic social etiquette and knowledge of group work and communication that will enhance their professional growth

COURSE OBJECTIVES:

- To understand what life skills are and their importance in leading a happy and well-adjusted life
- To motivate students to look within and create a better version of self.
- To introduce them to key concepts of values, life skills, and business communication
- To enable them to practice basic communication

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Recognize the need for life skills and values

CO-2: Recognize own strengths and opportunities

CO-3: Apply the life skills to different situations

CO-4: Understand the basic tenets of communication

CO-5: Apply the basic communication practices in different types of communication

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	-	1	2	1	1	2	2	3	3	3	2	-	-	-
CO-2	1	1	2	1	1	2	2	3	3	3	2	-	-	-
CO-3	1	1	2	1	1	2	2	3	3	3	2	-	1	-
CO-4	-	1	-	1	-	2	2	3	2	3	2	-	1	-
CO-5	-	1	-	1	-	2	2	3	2	3	2	-	1	-

LIST OF EXPERIMENTS:

UNIT-I:

Overview of Leadership-Oriented Learning:

- i) Self-Introduction
- ii) Recognise the need for life Skills and Values
- iii) Overview of Business Communication
- iv) Identify Strengths and Opportunities- Identity, body awareness
- v) Listening Skills: Law of nature- Importance of listening skills, Difference between listening and hearing; Types of listening.

UNIT-II:

Essential Grammar – I:

- i) Parts of speech & Tenses
- ii) Sentence Formation (General & technical)
- iii) Common errors
- iv) Voice
- v) Subject Verb Agreement

UNIT-III:

Overview of Communication Skills:

- i) Importance of effective communication
- ii) Types of communication: verbal and non-verbal
- iii) Barriers of communication, effective communication
- iv) Importance of Questioning
- v) Listening Skills: Law of nature- Importance of listening skills, Difference between listening and hearing; Types of listening.

UNIT-IV:

Written Communication:

- i) Letter Writing –Formal and Informal letter writing, Job application letter, E-Mail Writing
- ii) Summary writing
- iii) Story Writing
- iv) Report writing
- v) Building Curriculum Vitae.

UNIT-V:

Realities of Facing Life:

- i) Stress management working with rhythm and balance, and teamwork
- ii) Need for Life skills and values - importance, Critical life skills
- iii) Multiple Intelligences- Embracing diversity
- iv) Imparting Values: Leadership, dealing with ambiguity, motivation, creativity, and result orientation.

TEXT BOOK:

1. Handouts and reference links will be shared

REFERENCES:

1. Strategic Writing, Charles Marsh
2. The Seven Basic Plots, Christopher Booker
3. Business Communication, Saroj Hiremath
4. English Vocabulary in Use, Alan McCarthy and O'Dell

WEB REFERENCES:

Train your mind to perform under pressure- Simon Sinek <https://curiosity.com/videos/simon-sinek-on-training-your-mind-to-perform-under-pressure-capture-your-flag/>

Brilliant way one CEO rallied his team in the middle of layoffs [https://www. inc.com/video/simon-sinek-explains-why-you-should-put-people- before-numbers. html](https://www.inc.com/video/simon-sinek-explains-why-you-should-put-people-before-numbers.html)

Will Smith's Top Ten rules for success

[https://www. youtube. com/watch?v=bBsT9omTeh0](https://www.youtube.com/watch?v=bBsT9omTeh0)

ONLINE RESOURCES:

1. [https://www. coursera. org/learn/learning-how-to-learn](https://www.coursera.org/learn/learning-how-to-learn)
2. <https://www.coursera.org/specializations/effective-business-communication>

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25BS1MT107) LINEAR ALGEBRA

TEACHING SCHEME		
L	T/P	C
2	1	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Mathematical Knowledge at Pre-University level

COURSE OBJECTIVES:

- To learn concepts of Matrices, Rank of a matrix, Echelon form and solutions of system of linear equations
- To find Eigen Values, Eigen Vectors and application of Cayley-Hamilton theorem
- To learn concepts of complex matrices and conversion of quadratic form to canonical form
- To learn concepts of vector spaces such as independence, basis, dimensions and orthogonality
- To find Singular Value Decomposition, PCA and its applications

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Analyze the methods of finding the rank and solve a simultaneous linear system of equations

CO-2: Compute eigenvalues and eigenvectors to represent a linear transformation of matrices

CO-3: Reduce the quadratic form to its canonical form

CO-4: Understand the Rank, Basis and dimension to describe the matrices and subspaces

CO-5: Apply the knowledge of singular value decomposition in applications of image processing and machine learning

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	3	1	1	1	-	-	-	-	1	2	-	2	1
CO-2	2	3	3	2	1	1	-	-	1	1	2	2	2	1
CO-3	2	3	3	2	1	1	-	-	1	1	2	2	2	1
CO-4	3	2	3	2	1	2	3	-	1	1	2	1	2	-
CO-5	1	-	-	-	-	3	3	-	-	-	-	1	-	-

UNIT-I:

Matrices: Rank, Types of matrices, Linearly dependent and Independent vectors; Rank of a matrix by Echelon form and Normal form, Inverse of Non-singular matrices by Gauss-Jordan method; System of linear equations; Consistency of Homogeneous and Non-Homogeneous equations, LU Decomposition method.

UNIT-II:

Eigen Values and Eigen Vectors: Eigen values and eigenvectors and their properties, Diagonalization of square and symmetric matrices; Cayley-Hamilton Theorem (without proof), finding inverse and power of a matrix by Cayley-Hamilton Theorem.

UNIT-III:

Vector Space: Vector space; Subspace, linear independence, linear span, basis and rank, linear mappings.

UNIT-IV:

Inner Product Space: Norms, Inner product, metric space, orthogonality, orthogonal complement, orthogonal projections, orthogonal basis, Gram-Schmidt orthogonalization and QR decomposition

UNIT-V:

Applications: Singular value decomposition, Principal component analysis and Linear discriminant Analysis.

TEXT BOOK:

1. Higher Engineering Mathematics, B. S. Grewal, 42nd Edition, Khanna Publishers

REFERENCES:

1. Advanced Engineering Mathematics, Peter V. O'Neil, 7th Edition
2. Advanced Engineering Mathematics, Michael. D. Greenberg, 2nd Edition
3. Introduction to Linear Algebra, Gilbert Strang, 5th Edition
4. Applied Mathematics (Vol. I & II), P. N. Wartikar & J. N. Wartikar
5. Digital Image Processing, R. C. Gonzalez and R. E. Woods

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25BS1MT108) STATISTICAL MODELLING

TEACHING SCHEME		
L	T/P	C
2	1	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Mathematical Knowledge at Pre-University level

COURSE OBJECTIVES:

- To know sampling techniques
- To know methods of calculating correlation and regression
- To learn Methods of estimation
- To learn various methods to test the hypothesis
- To know basic concepts of Time series analysis

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Characterize the huge data using sampling techniques

CO-2: Calculate correlation, regression, rank correlation coefficients

CO-3: Estimate the parameters using different methods of estimation

CO-4: Apply the knowledge of different probability distributions to Test of Hypothesis

CO-5: Use Least squares method to compute time series

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	1	2	-	-	-	-	-	-	2	1	1
CO-2	3	3	2	1	2	-	-	-	-	-	-	2	2	2
CO-3	3	3	2	2	2	-	-	-	-	-	-	2	1	1
CO-4	3	2	2	1	2	-	-	-	-	-	-	2	1	1
CO-5	3	3	3	2	2	-	-	-	-	-	2	2	1	2

UNIT-I:

Sampling Techniques: Sampling, Types of sampling, Sampling from finite and infinite populations. Estimates and standard error (sampling with replacement and sampling without replacement), Sampling distribution of sample mean, stratified random sampling.

UNIT-II:

Linear Statistical Models: Simple linear regression & correlation, multiple linear regression & multiple correlation, Analysis of variance (one way, two-way).

UNIT-III:

Theory of Estimation: Point estimation, criteria for good estimates (un-biasedness, consistency), Methods of estimation including maximum likelihood estimation.

Sufficient Statistic: Concept & examples, complete sufficiency, their application in estimation.

UNIT-IV:

Inference:

Parametric Inference: Concept & formulation, Type I and Type II errors, Neyman Pearson lemma, Procedures of testing, large samples: Test concern single mean, difference of two means, single proportion, difference of two proportions.

Non-parametric Inference: Comparison with parametric inference, Sign test, Wilcoxon signed rank test, Mann-Whitney test, Run test, Kolmogorov-Smirnov test, Spearman's and Kendall's test

UNIT-V:

Basics of Time Series Analysis & Forecasting: Components of time series, Stationary, Autocorrelation function (ACF), Autocovariance function (ACVF)

Basic Time Series Models: AR, MA, ARMA and ARIMA Models, Identification, Estimation and Forecasting using Linear and quadratic trends.

TEXT BOOKS:

1. Probability and Statistics for Engineers, I. R. Miller, J. E. Freund and R. Johnson, 4th Edition
2. Introduction to Probability and Statistics for Engineers and Scientists, Sheldon M. Ross, 4th Edition, Academic Press, 2009
3. Fundamentals of Statistics (Vol. I & Vol. II), A. Goon, M. Gupta and B. Dasgupta.

REFERENCES:

1. Introduction to Linear Regression Analysis, D. C. Montgomery & E. Peck
2. Introduction to the Theory of Statistics, A. M. Mood, F. A. Graybill & D. C. Boes
3. Applied Regression Analysis, N. Draper & H. Smith
4. The Analysis of Time Series: An Introduction, Chris Chatfield

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25PC1CB101) DATA STRUCTURES AND ALGORITHMS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE OBJECTIVES:

- To introduce the fundamentals of algorithms, recursion, performance analysis, and coding practices
- To develop understanding of linear data structures and their applications
- To provide knowledge of tree-based non-linear data structures and their applications
- To impart knowledge of graph structures, algorithms, and their applications
- To introduce searching, sorting, and hashing techniques for efficient data handling

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Analyze algorithms using asymptotic notations and apply recursion with attention to time-space trade-offs

CO-2: Implement and apply arrays, stacks, queues, and linked lists for computational problem solving

CO-3: Construct and manipulate various trees (BST, AVL, B, B+ etc.) and use them in expression evaluation

CO-4: Implement graph algorithms (BFS, DFS, shortest path, spanning tree) and assess their efficiency

CO-5: Apply searching, sorting, and hashing techniques on data structures and evaluate their complexities

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	2	2	-	-	-	-	-	1	3	-	-
CO-2	3	3	2	2	2	-	-	-	-	-	1	3	-	-
CO-3	3	3	2	2	2	-	-	-	-	-	1	3	-	-
CO-4	3	3	2	2	2	-	-	-	-	-	1	3	2	-
CO-5	3	3	2	2	3	-	-	-	-	-	1	3	2	-

UNIT-I:

Basic Terminologies & Introduction to Algorithm and Data Organization: Algorithm specification, Recursion, Performance analysis, Asymptotic Notation - The Big-O, Omega and Theta notation, Programming style, refinement of coding-time-Space trade off, testing, data abstraction.

UNIT-II:

Linear Data Structure: Array, Table Lookup, Stack, Queue, Self-Referral Structure, Linked-list and its types, Various Representations, Operations & Applications of Linear Data Structures.

UNIT-III:

Non-Linear Data Structure: Trees (Binary Tree, Threaded Binary Tree, Binary Search Tree, B & B+ Tree, AVL Tree, Splay Tree). Applications: Expression Manipulation.

UNIT-IV:

Non-Linear Data Structure: Graphs (Directed, Undirected), Various Representations, Operations (search and traversal algorithms and complexity analysis) & Applications of Non- Linear Data Structures: Kruskals, Prims, Dijkstra's.

UNIT-V:

Searching and Sorting on Various Data Structures: Sequential Search, Binary Search, Insertion Sort, Selection Sort, Shell Sort, Divide and Conquer Sortings: Merge Sort, Quick Sort, Heap Sort, Introduction to Hashing.

TEXT BOOKS:

1. Fundamentals of Data Structures, E. Horowitz and S. Sahni, 1977
2. Data Structures and Algorithms, Alfred V. Aho, John E. Hopcroft, Jeffrey D. Ullman
3. Data Structures: A Pseudo-code Approach with C, Gilberg & Forouzan, Thomson Learning

REFERENCES:

1. The Art of Computer Programming: Volume 1: Fundamental Algorithms, Donald E. Knuth
2. Introduction to Algorithms, Thomas, H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein
3. Open Data Structures: An Introduction (Open Paths to Enriched Learning), Pat Morin, 31st Edition
4. An Introduction to Data Structures with Applications, Jean-Paul Tremblay & Paul G. Sorenson, Tata McGraw-Hill
5. Data Structures using C & C++, Ten Baum, Prentice-Hall International

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25ES1EC102) PRINCIPLES OF ELECTRONICS ENGINEERING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Fundamentals of Physics

COURSE OBJECTIVES:

- To understand the principle of operation and characteristics of various semiconductor devices
- To study the applications of various semiconductor devices
- To understand the operation and characteristics of Transistors
- To understand the concepts of feedback in amplifiers
- To know about analog and digital IC's

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand the principle of operation and characteristics of various semiconductor devices

CO-2: Understand the principles of operation of Diodes and its applications

CO-3: Comprehend the operation and characteristics of BJT and FET

CO-4: Analyse the effect of feedback in amplifiers and also understand the Op-Amp

CO-5: Understand the fundamentals of digital electronics

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	-	-	-	-	-	-	-	-	2	-	-
CO-2	3	2	1	-	-	-	-	-	-	-	-	3	-	-
CO-3	3	1	-	-	-	-	-	-	-	-	-	2	-	-
CO-4	3	2	-	-	-	-	-	-	-	-	-	2	-	-
CO-5	3	2	2	-	-	-	-	-	-	-	-	2	-	-

UNIT-I:

Diodes: Formation of P-N junction, forward and reverse biased P-N junction, V-I characteristics, Diode Equivalent circuit, Zener breakdown, Avalanche breakdown, Zener diode and its reverse characteristics.

Diode Applications: Operation and parameters of Half Wave Rectifier, Full Wave Rectifier

UNIT-II:

Bipolar Junction Transistors: Construction and Operation of NPN and PNP transistors, CE- CB- CC configurations, input and output characteristics.

Field Effect Transistors: JFET Structure and characteristics, MOSFET Structure and characteristics, depletion and enhancement type; CS, CG, CD configurations, CMOS basic principles.

UNIT-III:

Operational Amplifier Basics: Basic block diagram and pin diagram of Op-amp, ideal and practical Op-amp, inverting, non-inverting, and differential modes of operation, OP-AMP Applications: Adders, subtractors, V to I and I to V converters, Sample and Hold circuits, differentiators, Integrators.

UNIT-IV:

Digital Electronics Fundamentals: Boolean Algebra, Realization of Logic gates, Logic simplification using K- map

Combinational Circuits-half and full adder/subtractor, multiplexers, demultiplexers, Encoders and Decoders.

UNIT-V:

Sequential Circuits: Latches, Flip Flops-RS flip flop, D flip flop, JK flip flop, T flip flop, Shift Registers -SISO, SIPO, PISO, Asynchronous and Synchronous Counters.

TEXT BOOKS:

1. Millman's Integrated Electronics, Jacob Millman, Christos Halkias, Chetan Parikh, McGraw-Hill Education
2. Digital Logic & Computer Design, M. Morris Mano, Pearson
3. Linear Integrated Circuits, D. Roy Choudhary, Shail B. Jain, 5th Edition, New Age International, 2018

REFERENCES:

1. Electronic Devices & Circuit Theory, 11th Edition, Robert L. Boylestad, Louis Nashelsky
2. Solid State Electronic Devices, Sanjay Banerjee, 6th Edition, Ben Streetman
3. Microelectronics, Jacob Millman, Arvin Grabel
4. Electronics Devices & Circuits, S. Salivahanan, N. Suresh Kumar, A. Vallavaraj

ONLINE RESOURCES:

1. <https://engineerspower.com/nptel-online-courses-for-electrical-engineering/>
2. <https://mooc.global/tsinghua/fundamentals-of-electrical-engineering/>

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25HS1MG101) FUNDAMENTALS OF MANAGEMENT

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE OBJECTIVES:

- To understand the theories, functions and practices of management and to provide them with practical exposure to cases of success/failure in business
- To expose to a systematic and critical understanding of organisational structures, and design
- To explain key leadership styles, decision-making processes, and their impact on organisational effectiveness
- To comprehend the conceptual knowledge relating to Organisational Behavior and to provide a basic understanding of the behavior of individuals and groups in the organisations
- To develop an understanding of managerial ethics, corporate governance, and social responsibility in business practices

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Apply theories to improve the practice of management

CO-2: Describe and assess the basic design elements of organisational structure and design

CO-3: Apply leadership theories and decision-making models to real-world organizational problems

CO-4: Evaluate different models of organisational behavior and suggest strategies for managing stress and organisational change

CO-5: Evaluate ethical issues that organisations face in the fields of finance, sales, and marketing and demonstrate ethical reasoning and justify responsible business practices

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	-	-	-	-	-	1	-	-	1	1	2	-	1	3
CO-2	-	-	-	-	-	3	-	1	1	2	2	-	3	3
CO-3	-	-	-	-	-	2	1	3	3	2	2	-	3	3
CO-4	-	-	-	-	-	3	1	3	3	2	3	-	3	3
CO-5	-	-	-	-	-	3	3	2	3	2	2	-	2	3

UNIT-I:

Management Theories: Features of Management, Evolution of Management Thoughts [Pre-Scientific Management Era (before 1880), Classical Management Era (1880-1930, Neo-classical Management Era (1930-1950), Modern Management era (1950-on wards)].

Functions of Management: Planning: Concept, planning process, Types of plans;
Organising: Concept, nature, Principles of organisation;

Staffing: Nature and introduction to staffing functions;
Directing: Need for directing, principles and characteristics of directing;
Controlling: characteristics of control, steps

UNIT-II:

Organisational Structures: Organisation chart, Organisational structure - Line Organisation, Line and Staff Organisation, Functional Organisation, Committee Organisation, Divisional Structure, Matrix Structure, Boundaryless Structure, Virtual Structure, and Inverted Pyramid Structure.

(Definition, Structure, merits, demerits and suitability for all organisation structures).

UNIT-III:

Leadership: Concept, Nature, Types, Importance, Attributes of a Leader, Leadership Grid.

Decision Making: Concept and process, types of Decision-making.

UNIT-IV:

Organisational Behavior: Definition, Approaches- Human Resource, Contingency, Productivity, and Systems approach; Models- Autocratic, Custodial, Supportive, Collegial, and System Model.

Personality: Definition; Determinants.

Perception: Definition, Factors Influencing Perception.

Stress Management: Definition, Types, Consequences, and Strategies to Manage

Organisational Change: Definition; Resistance; Overcoming and Approaches: Lewin's Three-Step Change Model, Kotter's Eight-Step Plan for Implementing Change.

Organisational Culture: Definition; Functions; Factors influencing.

UNIT-V:

Managerial Ethics: Ethics and Business: Definition, Need and Types, Ethics of Marketing & Advertising, Ethics of Finance & Accounting, Corporate Governance: Definition, Benefits, How to achieve good Governance, Corporate Social Responsibility: Definition, Types, Benefits, Social Responsibility of Business.

TEXT BOOKS:

1. Understanding the Theory and Design of Organizations by Richard L. Daft, 11th Edition, Cengage, 2020
2. Management, James Arthur, Finch Stoner, R. Edward Freeman, and Daniel R. Gilbert, 6th Edition, Pearson Education/Prentice Hall
3. Organizational Behaviour, Stephen P. Robbins, Timothy A. Judge, Pearson Education, 2019

REFERENCES:

1. Fundamentals of Management: Essential Concepts and Applications, 6th Edition, Pearson Education, 2009
2. Fundamentals of Management, Amanjot Sachdeva & Dr. Pardeep Kumar, S. Chand, 2012
3. Fundamentals of Management: Essential Concepts and Applications, Robbins S. P., DeCenzo D. A, Prentice Hall, 1995
4. Fundamentals of Management, Pathak J. P., Vikas, 2015
5. Principles of Management: Functions and Fundamentals of Management, Prabhu Thankaraju, 2020

Effective

B. Tech. II Semester

(25PC2CB101) DATA STRUCTURES AND ALGORITHMS LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To understand and apply linear data structures (arrays, stacks, queues, linked lists) to solve computational problems
- To implement non-linear data structures (trees, graphs) and their applications in real-world scenarios
- To develop problem-solving skills using recursive and iterative approaches with suitable data structures
- To analyze the efficiency of algorithms and programs through time complexity analysis
- To demonstrate practical skills in storing, retrieving, and manipulating data structures using file handling

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Implement and analyze algorithms using arrays, stacks, queues, and linked lists for various applications

CO-2: Apply stack-based and queue-based techniques to solve expression evaluation, recursion, and priority-based problems

CO-3: Construct and manipulate different types of trees (BST, AVL, B-Trees, B+ Trees, Threaded Trees) and implement expression trees

CO-4: Implement graph algorithms (BFS, DFS, shortest path, spanning tree) and evaluate their performance

CO-5: Develop programs to store and retrieve non-linear data structures from files and perform complexity analysis

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	2	2	-	-	-	-	-	1	3	-	-
CO-2	3	3	2	2	2	-	-	-	-	-	1	3	-	-
CO-3	3	3	2	2	2	-	-	-	-	-	1	3	2	-
CO-4	3	3	2	2	2	-	-	-	-	-	1	3	2	-
CO-5	3	3	2	2	3	-	-	-	-	-	1	3	2	-

LIST OF EXPERIMENTS:

WEEK 1:

Implement applications using arrays, Line editors with line count, word count showing on the screen

WEEK 2 & 3:

Implement Stacks using arrays and linked list

Stacks Applications: Recursion problems, Expression Manipulation, Towers of Hanoi using user defined stacks.

WEEK 4 & 5:

Implement various types of queues using arrays and linked list

Queues Applications: Priority Queues

WEEK 6 & 7:

Implement various types of linked list, perform operations: insert, delete, merge

Linked List Applications: polynomial manipulation

WEEK 8 & 9:

Implement insert, delete, traverse operations on Binary Trees, BST, B Trees, B+ Trees, AVL Trees, Threaded Binary Trees.

WEEK 10 & 11:

Expression Trees.

WEEK 12 & 13:

All graph algorithms.

WEEK 14 & 15:

Saving / retrieving non-linear data structure in/from a file. Sorting

TEXT BOOKS:

1. Fundamentals of Data Structures, E. Horowitz and S. Sahni, 1977
2. Data Structures and Algorithms, Alfred V. Aho, John E. Hopcroft, Jeffrey D. Ullman
3. Data Structures: A Pseudo-code Approach with C, Gilberg & Forouzan, Thomson Learning

REFERENCES:

1. The Art of Computer Programming: Volume 1: Fundamental Algorithms, Donald E. Knuth
2. Introduction to Algorithms, Thomas, H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein
3. Open Data Structures: An Introduction (Open Paths to Enriched Learning), Pat Morin, 31st Edition
4. An Introduction to Data Structures with Applications, Jean-Paul Tremblay & Paul G. Sorenson, Tata McGraw-Hill
5. Data Structures using C & C++, Ten Baum, Prentice-Hall International

B. Tech. II Semester

(25ES2EC102) PRINCIPLES OF ELECTRONICS ENGINEERING LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To know the characteristics of various semiconductor devices
- To verify the applications of various semiconductor devices
- To understand the operation and characteristics of Transistors
- To verify the functionality and applications of analog IC's
- To verify the functionality of digital IC's

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Analyze the characteristics of diodes

CO-2: Interpret the characteristics of transistors

CO-3: Apply the knowledge of semiconductor devices

CO-4: Understand the functionality of Op-amp and its applications

CO-5: Comprehend the functionality of digital IC's

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	-	-	-	-	-	-	3	1	2	-	-	-
CO-2	3	2	-	-	-	-	-	-	3	1	2	-	-	-
CO-3	3	2	-	-	-	-	-	-	3	1	2	-	-	-
CO-4	3	2	-	-	-	-	-	-	3	1	2	-	-	-
CO-5	3	1	1	-	-	-	-	-	3	1	2	-	-	-

LIST OF EXPERIMENTS:

1. Forward and Reverse Bias V-I characteristics of PN junction Diode.
2. V-I characteristics of Zener diode.
3. Full wave rectifier.
4. Characteristics of a BJT under CB configuration.
5. Characteristics of a BJT under CE configuration.
6. JFET characteristics under CS configuration.
7. MOSFET characteristics under CS configuration.
8. Inverting and Non-Inverting amplifiers using IC 741 Op-Amp.
9. Adder and subtractor using IC 741 Op-Amp.
10. Integrator and Differentiator using IC 741 Op-Amp.
11. Truth table verification of Logic gates.
12. Truth table verification of Half-Adder and Full Adder

B. Tech. II Semester

(25PC2CB102) PYTHON PROGRAMMING

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To learn the fundamentals of Python programming and data structures
- To perform input/output operations on different data structures
- To use standard Python libraries (NumPy, Pandas, Matplotlib) for solving problems
- To apply basic statistical and data analysis methods to real-world datasets

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand and implement various data structures in Python

CO-2: Apply predefined Python libraries (NumPy, Pandas) for data processing applications

CO-3: Analyze datasets using basic statistical methods and simple regression

CO-4: Visualize data by plotting different types of graphs using Matplotlib

CO-5: Know the importance of data analytics and its applications in real-world applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	1	1	-	1	1	-	-	-	-	-	1	2	-	-
CO-2	1	1	-	1	1	-	-	-	-	-	1	2	-	-
CO-3	1	1	1	1	1	-	-	-	-	-	1	2	-	-
CO-4	1	1	-	1	1	-	-	-	-	-	1	2	-	-
CO-5	1	1	1	1	1	-	-	-	-	-	1	2	-	-

LIST OF EXPERIMENTS:

WEEK 1, 2, 3,4,5,6:

Python Concepts, Data Structures, Classes: Interpreter, Program Execution, Statements, Expressions, Flow Controls, Functions, Numeric Types, Sequences, lists, dictionaries, Text & Binary Files - Reading and Writing

WEEK 7, 8, 9, and 10:

Pandas, Numpy library

Visualization in Python: Matplotlib package, Plotting Graphs, Controlling Graph, Adding Text, More Graph Types, Getting and setting values

WEEK 11:

Demonstrate simple linear regression analysis. Analyze results in detail.

WEEK 12:

Demonstrate multiple regression model. Analyze results in detail.

WEEK 13:

Demonstrate Logistic regression model. Analyze results in detail. 40

WEEK 14:

Demonstrate other regression models. Compare results in detail 11,12,13

TEXT BOOKS:

1. Introduction to Linear Regression Analysis, D. C. Montgomery and E. A. Peck, 3rd Edition, Wiley, 2006
2. Programming Python, Mark Lutz, 4th Edition, O'Reilly Media, 2011
3. Introduction to Statistical Analysis of Laboratory Data, Bartolucci A., Singh K. P., Bae S., Wiley, 2015

REFERENCES:

1. Regression Diagnostics, Identifying Influential Data and Sources of Collinearity, D. A. Belsey, E. Kuh and R. E. Welsch, Wiley-Interscience, 2013
2. Python 3 for Absolute Beginners, Tim Hall and J. P. Stacey, Apress, 2009
3. Python for Data Analysis, Wes McKinney, 3rd Edition, O'Reilly Media, 2022

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25HS2EN103) BUSINESS COMMUNICATION AND VALUE SCIENCE - II

TEACHING SCHEME		
L	T/P	C
0	2	2

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES:

- Basic Knowledge of English (verbal and written), Completion of all units from Semester 1

COURSE OBJECTIVES:

- To develop effective writing, reading, presentation and group discussion skills
- To identify personality traits and evolve as a better team player
- To introduce key concepts of Morality, Behavior and beliefs, Diversity & Inclusion

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Use tools of structured written communication and electronic/social media to share concepts and ideas

CO-2: Understand the basics of presentation and apply effective techniques to make presentations

CO-3: Understand and apply the basic concept of speed reading, skimming and scanning

CO-4: Identify individual personality types and role in a team and recognize the concepts of outward behavior and internal behavior

CO-5: Understand the basic concepts of Morality and Diversity

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	-	1	1	1	1	2	2	2	2	3	2	-	1	-
CO-2	1	1	2	1	1	2	2	2	2	3	3	-	1	-
CO-3	1	1	2	2	1	2	2	2	1	3	2	-	1	-
CO-4	-	1	1	1	-	2	2	2	3	2	2	-	-	-
CO-5	-	1	-	1	-	2	2	3	2	3	2	-	-	-

LIST OF EXPERIMENTS:

UNIT-I:

Team Building:

- Introduction of Dr. Meredith Belbin and his research on team work
- Belbin's 8 Team Roles
- Lindgren's Big 5 personality traits

- iv) Team Falcon Practical to identify individual personality traits with Belbin's 8 team player styles
- v) Build a team for NGO

UNIT-II:

Essential Grammar and Reading Skills:

- i) Good and Bad Writing -Discussion
- ii) Common errors, Punctuation rules
- iii) Lucid writing – writing techniques
- iv) Speed Reading
- v) Skimming and Scanning

UNIT-III:

Presentation and Social Media Skills-1:

- i) Prepared Presentation
- ii) Design a street play and promote the play through social media
- iii) Prepare and Publish E-magazine
- iv) Use electronic/social media to launch the E-magazine
- v) Capture the reviews and likes

UNIT-IV

Branding & Satori:

- i) Importance of Branding- Types of Branding-Digital branding
- ii) Sharing the learning with peers (Satori)
- iii) Branding: NGO and E-Magazine.

Diversity:

- iv) Diversity & Inclusion- Different forms of diversity in our society
- v) Ethics, Morality and respect for Individual – Diversity.
- vi) Challenges faced by members of diverse groups.

UNIT-V:

Creation of an organization - Mock NGO:

- i) Three A's – Aware, Articulate and Amplify
- ii) Individual identification of Social Issues
- iii) Materials to create an identity of an organization –Vision, Mission, Values,
- iv) Research on the Social cause and generate the Importance of Integrity, Responsibility, Excellence, Unity & Pioneering
- v) Report writing

TEXT BOOK:

1. There are no prescribed texts for semester II – there will be handouts and reference links shared.

REFERENCES:

1. Guiding Souls: Dialogues on the Purpose of Life; Dr. A. P. J. Abdul Kalam, Arun Tiwari, 2005
2. The Family and the Nation; Dr. A. P. J. Abdul Kalam, Acharya Mahapragya, 2015

3. The Scientific India: A Twenty First Century Guide to the World Around Us; Dr. A. P. J. Abdul Kalam, Y. S. Rajan, 2011
4. Forge Your Future: Candid, Forthright, Inspiring; Dr. A. P. J. Abdul Kalam, 2014

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25MN6HS102) ENVIRONMENTAL SCIENCE

TEACHING SCHEME		
L	T/P	C
2	0	0

EVALUATION SCHEME		
SE-I	SE-II	TOTAL
50	50	100

COURSE PRE-REQUISITES: Basic Knowledge on environmental issues

COURSE OBJECTIVES:

- To recognize the principles of environment
- To identify the sources and causes of Environmental Pollution
- To analyze the causes of climate change and identify climate change mitigation and adaptation technologies
- To list out the benefits in creating circular economy and sustainability
- To emphasize the Environmental conventions, policies and regulations

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Gain knowledge about principles of Environment and allied problems

CO-2: Identify the causes and suggest remedial measures for Environmental Pollution

CO-3: Appraise the mitigation measures and adaptation technologies related to climate change

CO-4: Familiarize with the importance of circular economy towards sustainability

CO-5: Relate and implement environmental policies and regulations

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	2	2	1	-	1	2	2	1	1	1	2	-	-	-
CO-2	2	2	1	-	1	2	2	1	1	1	2	-	-	-
CO-3	2	2	1	-	1	2	2	1	1	1	2	-	-	-
CO-4	2	2	1	-	1	2	2	1	1	1	2	-	-	-
CO-5	1	1	1	-	-	2	1	1	1	1	-	-	-	-

MODULE 1:

Introduction to Environmental Science: Importance and Scope of Environmental Science. Overview of Environment & its components. Ecosystem function and uses. Biodiversity – Types and Values.

MODULE 2:

Synergy With Environment: Relationship between environmental science & society – Influence of Industry & infrastructure on environment (Specific case studies). Sources, Impacts and prevention of air, water and soil pollution.

MODULE 3:

Climate Change: Science behind climate change – factors responsible for climate change, Role of greenhouse gases – Global temperature rise & its impact on environment & human health. Carbon footprint – Briefing on Paris agreement, Identify

key sectors for low carbon footprint. Climate change mitigation & adaptation strategies.

MODULE 4:

Moving Towards Sustainability: EIA and EMP - importance, Sustainable agriculture – Importance of Organic farming and hydroponics. Role of AI & IOT for efficient management of environmental issues – Health, air, water, and soil. Sustainable living practices – Principles and methods (Linear Economy vs circular economy). Green Buildings.

MODULE 5:

Policies and Regulations: International Conventions: Montreal Protocol, COP15, Kyoto protocol (Salient Features). National Environmental Policies – Salient features of EPA act, Air act, Water Act, Forest and Wildlife act. National Green Tribunal Act (2010)- Features and Significance. Extended producer responsibility. Innovative approaches to waste management: Plastic, tyre and batteries.

TEXT BOOKS:

1. Living in the Environment, G. Tyler Miller/ Scott E. Spoolman, 20th Edition, Cengage Learning
2. Environmental Studies, Rajagopalan, Oxford University Press, 2005
3. Introduction to Climate Change, Andreas Schmittner, Oregon State University, 2018. <https://open.oregonstate.edu/education/climatechange/>

REFERENCES:

1. Green Development: Environment and Sustainability in a Developing World, Bill Adams, 4th Edition, Routledge Publishers, 2021
2. Fixing Climate, Robert Kunzig & Wallace S. Broecker, Profile Books Publisher, 2009
3. Plastic Waste and Recycling-Environmental Impact, Societal Issues, Prevention and Solutions, 1st Edition, Edited by Trevor Letcher, Academic Press, 2020
4. Waste Management Practices: Municipal, Hazardous, and Industrial, John Pichtel, CRC Press, 2020
5. Environmental Science: A Global Concern, William P. Cunningham, Mary Ann Cunningham, McGraw-Hill Education, 2022

REFERENCES:

1. https://www.google.co.in/books/edition/Waste_Management_Practices/pG3SBQAAQBAJ?hl=en&gbpv=1&pg=PA1&printsec=frontcover